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NAME: Bhuvanesh B STD.: _____ SEC.: _____ ROLL NO.: _____ SUB.: _____

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GM 16.9.26

21.07.26 Ex no: 01

Network Commands

Aim:

To study various network Commands used in Linux and windows

Procedure:

1. open the Command prompt Terminal
2. Execute the sequenced network Commands
3. Observe and record the output of each Commands
4. Analyse the network Information display by the Commands

COMMANDS:

1. ARP Command

~ arp -a

purpose: Display IP and MAC address information

2. Hostname:

~ hostname.

purpose: Display name of the Computer

3. IP Configuration

~ ipconfig /all

purpose: Display details TCP/IP Configuration information

4. netBIOS

~ netstat -a

purpose: Display network Connections and routing info

5. network Statistics:

~ netstat -r

purpose: Displays net BIOS name information

6. Dns lookup:

~nslookup www.google.com

purpose : Display Dns information if the Specified domain

7. Ping

~Ping google.com

purpose: Checks Connectivity between the local System and a destination

8. Route

~route

purpose: Displays or modifies the IP routing table

9. IP Command

~IP address show

purpose: Display IP address assigned to network interface

10. MTR:

~mtr google.com

purpose : Displays the router and network statistics for each hop

11. TCP Dump:

~tcpdump -i eth0

purpose : Captures and displays network packets

12. nmcli:

~nmcli Connection show

purpose : Display network manager Connection profiles

Sample output:

C:/> arp -a
Interface : 192.168.1.10

Internet Address	Physical address	Type
192.168.1.1	00-11-22-33-44-55	dynamic

C:/hostname
Desktop-PC

C:/> ipconfig /all
IPv4 address : 192.168.1.10
Default Gateway : 192.168.1.1
DNS Server : 8.8.8.8
C:/>net stat -r

network destination	gateway
0.0.0.0	192.168.1.1

C:/> nslookup www.google.com
name : www.google.com
Address : 142.250.195.68
C:/> Ping google.com
Reply from 142.250.195.68 : bytes = 32 time = 20ms TTL=117
Reply from 142.250.195.68 bytes = 32 time = 18 ms TTL = 117.

Result:

Thus, the command to run network Configuration, IP address, MAC address, routing information, DNS details, and Connectivity status were displayed Successfully.

Ex no: 2

STUDY OF NETWORK CABLES.

Aim:

To study the different type of network Cables and to make Straight - through and Crossover ethernet cables using a tool.

Procedure:

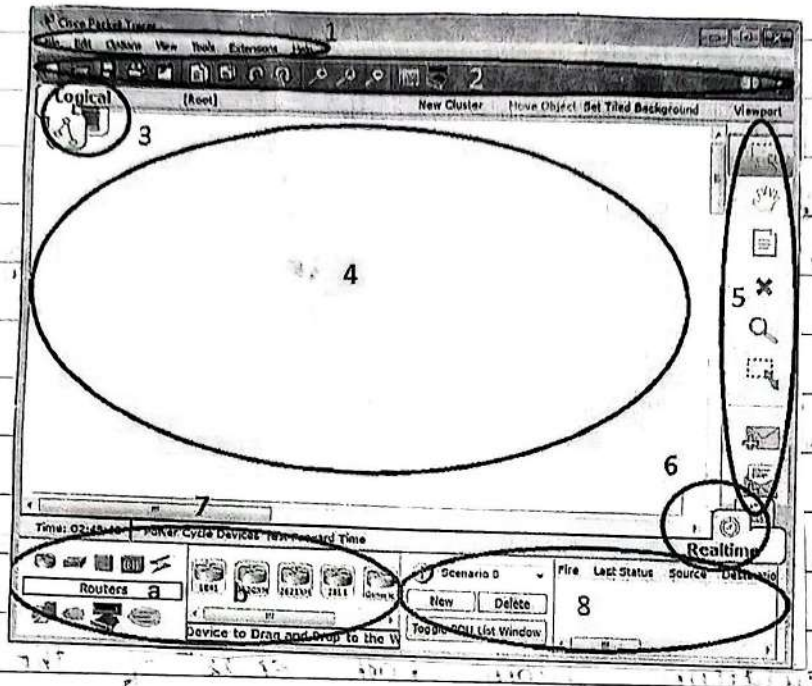
1. Take the required CAT5/CAT5e ethernet cable, crimping tool and two RJ45 Connectors
2. Strip approximately 1.5cm of the outer insulation from both ends of the cable.
3. Untwist the four pairs of wires and arrange them in the required colour sequence
4. For a straight - through cable, use the same wiring arrangement at both ends. For B at the corner.
5. Align the wires evenly and Insert them carefully into RJ45 Connector without Changing their order
6. Insert the RJ45 Connector without Changing the Crimping tool and Crimp at firmly
7. Repeat the procedure for the other and test the Cable by Connecting network.

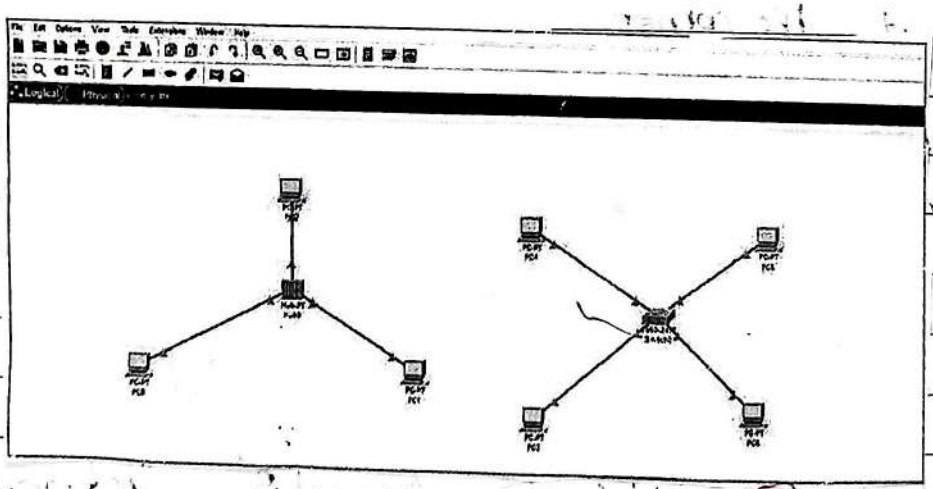
18/8 9/10

Result:

Thus the different types of network Cables were studied and the Straight - through and Crossover ethernet cable.

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Ex - 03

PACKET TRACKER TOOL INSTALLATION AND USER INTERFACE OVERVIEW.

Aim:

To understand the Installation and User interface of Cisco packet Tracer and to design a simple network using network device.

Introduction:

Cisco packet tracer is a network simulation and modelling tool, developed by Cisco it allow user to Create configure and test network without using physical hardware.

User Interface:

The main part of packet Tracer are.

1) menu bar - Used for file operations
2) main Toolbar - Provides shortcuts such as open, Save, Undo, redo.
3) Logical / Physical workshop - Used to switch between logical and Physical
4) Workspace - Area where network topologies are created
5) Common Tools Bar - Provides tools such as select
6) Realtime - Used to view normal network
7) network Component Box - Contains routers Switches.
8) pack box - Used to Created and Observe.

Procedure :

1) Create Network Using hub.

Open Packet tracer
places 4 PCs and 1 Hub in workspace.
Select Connection -> copper Straight

2) Create Network Using Switch.
place 4 PCs and 1 switch
Send a PDU from one Pc to another
Observe the packet movement in simulation.

Observation :

Send the received frame to all Connected ports

Does not identify the destination device
Create more Unnecessary traffic

Topology :

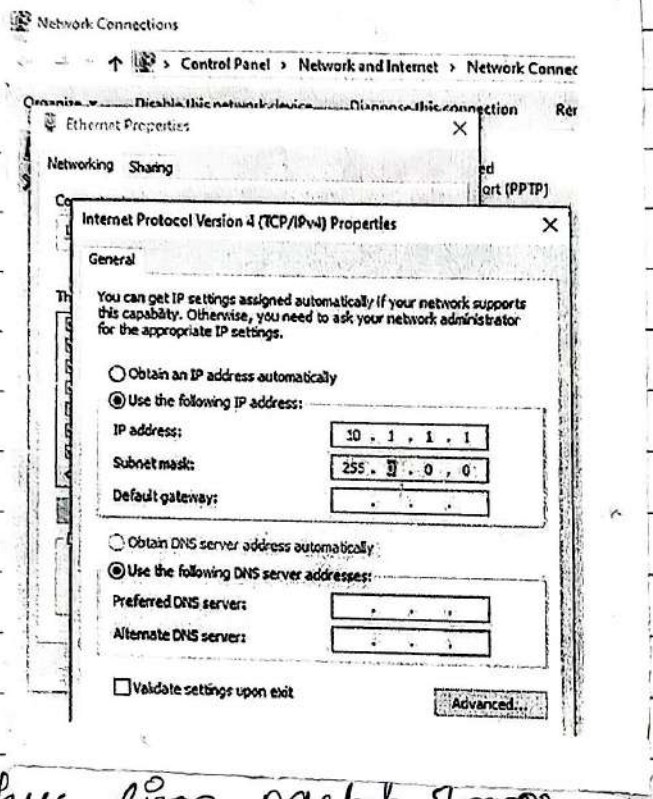
The network Create Using a hub/switch with multiple PCs represent Star Topology because all devices are Connected to a Central device

Switch:

forward the frame mainly to the destination part

Reduces Unnecessary traffic

mainly work at the Data link Layer



Result:

Thus Cisco packet Tracer was studied and a simple network using PCs, hub and switch was Successfully designed and Tested.

9/10

Ex-5

WIRESHARK

Aim: Experiments on packet capture tool wireshark.

To filter, capture, View, Packets in wireless Tool capture 100 packet from the Ethernet, IEEE 802.3 LAN Interface Save it

procedure:

Select local area connection in wireless
To capture option.
Select stop capture automatically after 100 packets.
Then click start capture
Search TCP packet in reach box
To see flow graph click statistic flow graph.
Save the packets.

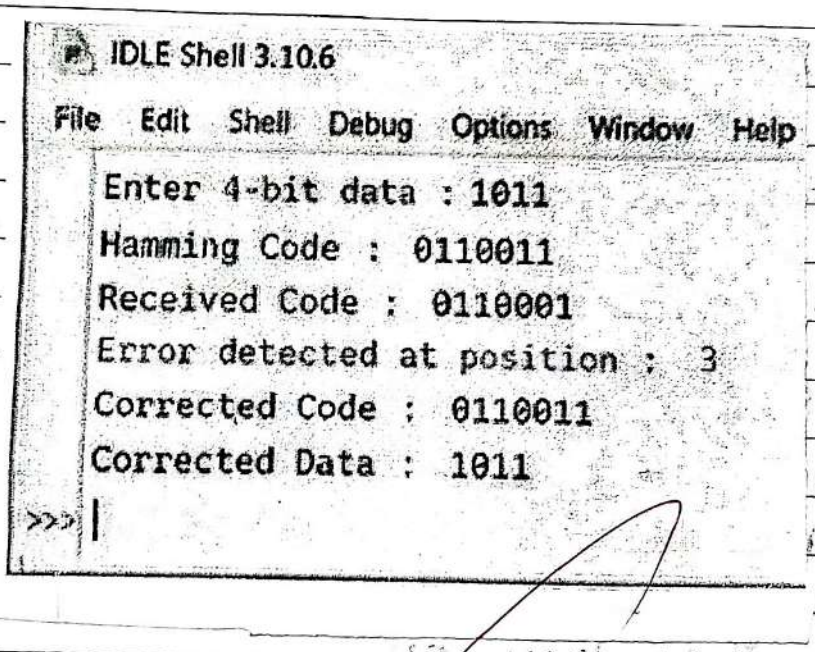
1) Create a filter to display only TCP/UDP packets, inspects the packets and provide the flow graph.

Select LAN connection in wiresharks
To Capture option
Select stop capture automatically after 100 packets
Then click start Capture.
Save the packets.

Step 4:- Check the connectivity between switch and other machine by using ping command in the command prompt.

Step 5:

Try to access the shared folder form other computers of the network.



Result:

Wireless LAN using switch and ethernet cables are executed successfully in packet tracer.

Ex - 4

Setup and Configure a LAN, using Switch and Ethernet cables.

What is LAN?

Lan stands for local area network which is a group of computers and connected devices linked together in a small physical location like a home, school or office building

How to set up LAN:

Step-1 - Connect the hardware.

-> Place your router in a central location near a power outlet.

-> Connect your modem to the router's WAN or Internet port using an Ethernet cable

-> plug the router into power and wait for it to turn on.

Step 2 - Connect your devices

-> wired setup: plug one end of an Ethernet cable into your computer network.

Step 3 - Configure the router.

-> Type your IP address (192.168.1.1 or 192.168.0.1) into address bar and press Enter.

Ex-6

ERROR DETECTION & CORRECTION USING HAMMING CODE

Aim:

Write a program to implement error detection and Correction using hamming code Concept make a test run to input data stream and verify

Sender:

Input to sender file should be a text of any length program should be concert the text to binary

Apply hamming code concept on the binary data and redundant bits to it

Code

```
data = input ("Enter 4 bits:")
d1, d2, d3, d4 = map (int, data)
P1 = d1 ^ d2 ^ d4
P2 = d1 ^ d3 ^ d4
P4 = d2 ^ d3 ^ d4

Code = [P1, P2, d1, P4, d2, d3, d4]

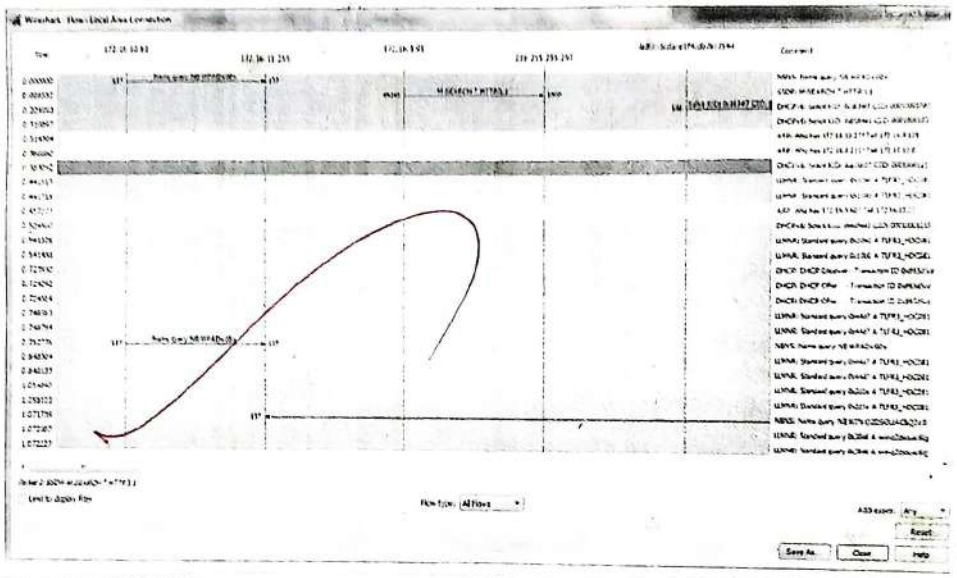
Print ("Sender Hamming Code:",
Join (map (str, code)).
```

Receiver Side:

Receive the hamming code from the Sender.

5) Create a filter to display only DHCP packets and inspect the packets.

Select local area Connection in Wireshark
Go to capture option
Select stop capture automatically after 100 packets
Then click Start capture.
Save the packets.



Result:

Wireshark on packet capture tool experiment are learned and executed successfully.



2) Create a filter to display only ARP packets and inspect the packets.

capture option.

select stop capture automatically after 100 packets,

then click start capture

Save the packets

3) Create a filter to display only HTTP packets and inspect the packets procedure

-> Select local area Connection in wireshark

Go to capture option

Select Stop capture automatically after 100 packets.

Search HTTP packets in search bar

4) Create a file to display only DNS packets and provide the flow graph

Go to capture option

select Stop capture automatically after 100 packets

then clicks start capture

Search DNS packets in search bar

To see the flow graph click Statistics. flow growth.

Ex-07.

SLIDING WINDOW PROTOCOL:

Aim:

Write a Program to implement how control of date link Layer, Using sliding window protocal. program should archive at least below given requirements you can make it a bidirectional progeram.

Create a sender program with following

1) Input window size from the user.
2) Input a Test message from the User.
3) Consider 1 character frame
4) Create a frame with following fiels.
5) send the frames
6) wait for the acknowledgement from the Reciver

procedure:

7) Task 1 - Introduction,

=> AKP Scan: this scan uses AKP request. to discover

Code:

```
import time
window = int (input ("Enter window size:"))
message = input ("Enter message")
start = 0
while start < len (message)
print ("SENDER")
end = start + window
if end > len (message):
    end = len (message)
```

7. Sliding Window Protocol (Flow Control)

```
Enter window size : 3
Total number of frames : 7

Sending frames...
Frame 0 sent        ACK received
Frame 1 sent        ACK received
Frame 2 sent        ACK lost  ->  Retransmitting
Frame 2 sent        ACK received
Frame 3 sent        ACK received
Frame 4 sent        ACK received
Frame 5 sent        ACK received
Frame 6 sent        ACK received

Transmission completed successfully.
```

Result :

Thus a Error Detection & correction Hamming Code is in implemented successfully.

Enter the received code
Calculate the parity code
Find the error position using the parity values
Display the Corrected hamming code

Code:

```
r= list (map (int, input ("Enter received
                    -t bits"))
r1 = [0] ^ r[2] ^ r[4] ^ r[6]
r2 = r[1] ^ r[2] ^ r[5] ^ r[6]
r3 = r[3] ^ r[4] ^ r[5] ^ r[6]

error = r1*1 + r2*2 + r4*4
if error = 0
    print ("no error")
else:
    print ("Error at position:", error)
    r[error-1] ^= 1
print ("Corrected code:", "".join(map
        (str, received)))
```

Input

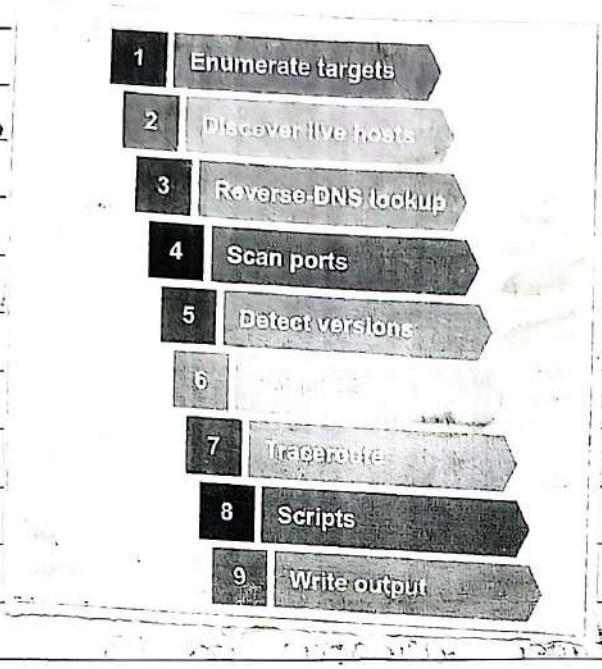
Enter 4 data bits : 1011
Sender : Hamming Code : 0b1100 11
Receiver : 7 bits : 0010011

Output

Error at position : 2
Corrected code : 0110011

Sample I/P - 2

Enter 4 data bits : 0010



Result:

Sliding window protocols are learned and executed successfully.

Sender reads Ack / NAck.

```
print ("In Sender (Hecking Response")
response = receiver_buffer[0]
number   = receiver_buffer[1]
if response = Ack:
    print ("Ack received:", number)
    start  = number+1
else:
    print ("NACK received", number)
    start = number
print ("Transmission Completed").
```

output

Enter window size : 3
Enter message : Hello.

Sender:
Frame NO : 0 DATA: H
Frame NO : 1 DATA: e
Frame NO : 2 DATA: l

SENT

RECEIVER
Received Frame: 0 DATA: H
Received Frame: 0 DATA: e
Received Frame: 2 DATA: l.

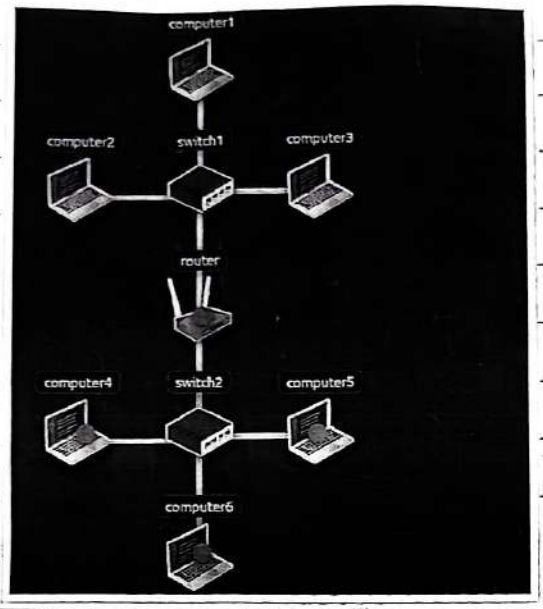
Received buffer: ('Ack', 2)
Ack Received: 2.

Transmission Completed.

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```
# sender buffer

Sender_buffer = []
for i in range (start, end):
    frame = buffer.append(frame)
    sender_buffer
    print("\n sender buffer :")
    print(sender_buffer)
# Delay
    print("\n waiting for receiver...")
    time.sleep()
# Receiver
    print("Receiver")
    expected = start
    error = false
for frame in sender_buffer:
frame_no = frame[0]
 data = frame[1]
 print("Received frame:", frame_no, "DATA:", data)
 if frame_no = expected:
    else:
        error = True
        break
# receiver buffer,
if error:
    receiver_buffer = ("nack", expected)
    print("\n Receiver Buffer:")
    print(receiver buffer)
else:
    ack = expected + 1
    receiver_buffer = ("Ack", ack)
    print("\n. Receiverbuffer")
    print("receiver buffer")
```



How many devices can see the ARP request -
4

Did computer 1 reply to the ARP request
Y

Enumerating Targets

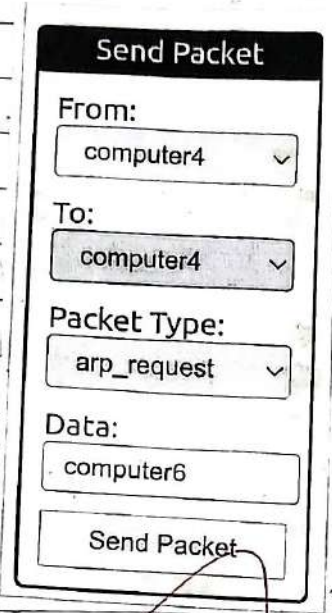
Before delving into details of scanning techniques. It's essential to define the target we want to scan Target can be specified in three ways.

1) List you can provide a list of specific IP address

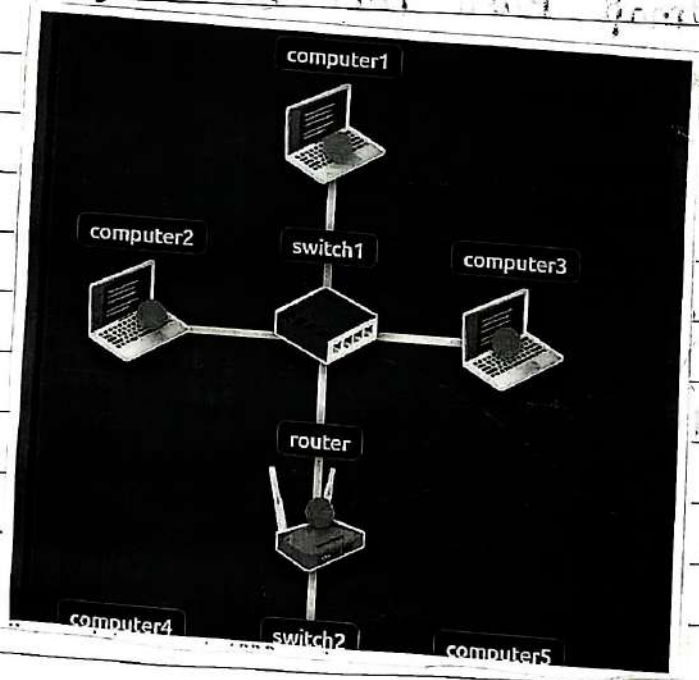
2) Range: you can specify a rang such as 10.11.12.15-20

3) Subnet - you can define a subnet like "machine_IP/30,

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from Computer 1
To Computer (to indicate)
packet type: "ARP Request"
Computer 6



How many devices can see the ARP request
Did Computer 6 receive the ARP Request?

Ex: 8.

ARP, ICM, TCP/UDP.

Aim:

To Discover live hosts Using Scans(ARP, ICMP, TCP/UDP) on the Try hackme platform room

Introduction:

When targetting a network, we need an efficient tool to handle repetitive task, this tool should help us find out which System are online and services are running on those system. The tool that we will rely on Nmap. The first Question about finding live Computer is answered in the room

Nmap live host Discovery
Nmap Basic Port Scans
Nmap Advanced port scans

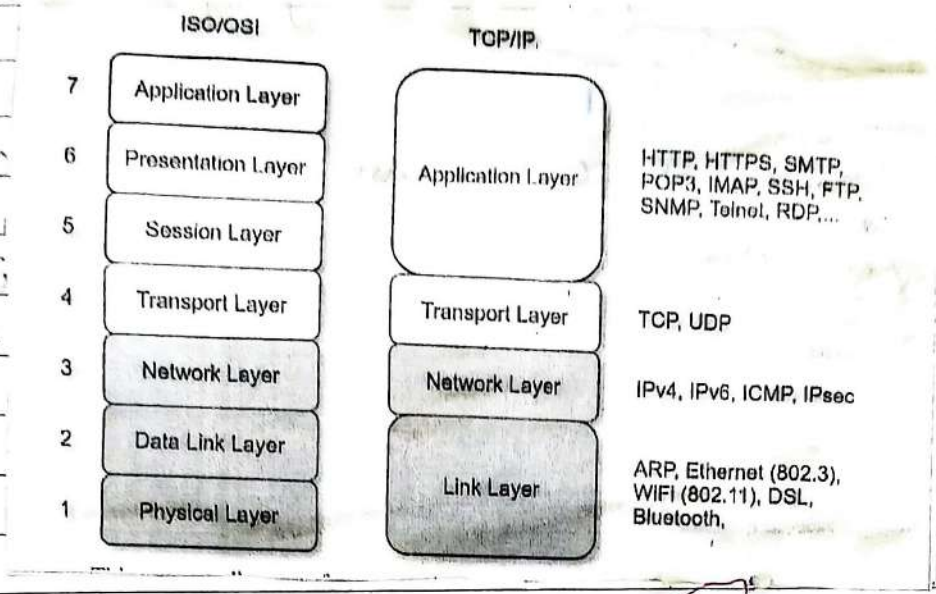
ARP scans - this scan use ARP request to discover live hosts

ICMP scans - this Scan uses ICMP requests to identify live hosts

TCP/UDP ping Scan - this Scan sends packets to TCP ports and UDP ports to determine live hosts.

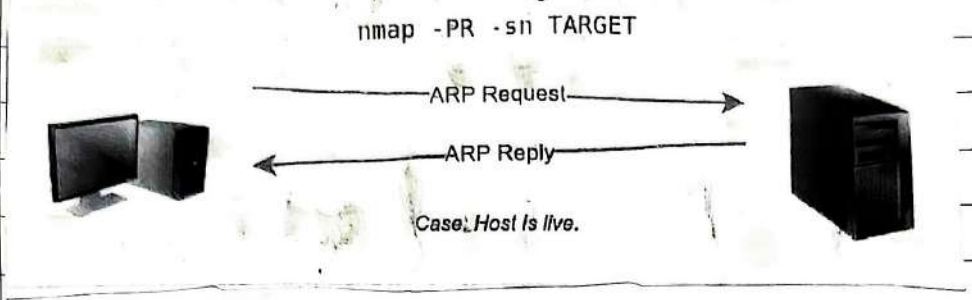
Subnetwork:

A network segment is a group of Computer linked through a shared medium like an Ethernet switch or WIFI access point.

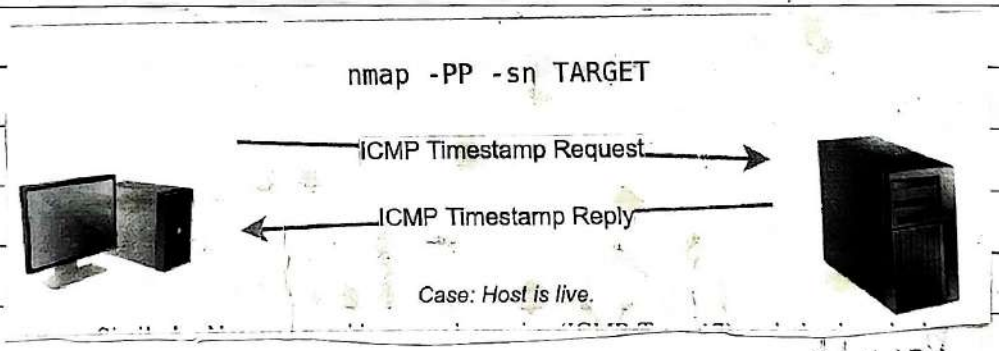


Result:

We have learned how ARP, ICMP, TCP and UDP can detective live hosts by Completing this room.



Nmap uses timestamp request (ICMP type 13) and checks whether it will get a Timestamp reply (ICMP type 14) adding the PP option tells nmap to use ICMP timestamp request.



This passage discusses four network protocols ARP, ICMP, TCP and UDP and their role in network scanning. ARP.

MAC address with a specific IP on the same subnet an ARP query should be sent before ICMPE echo.

Additional network scanners can use specially-crafted packets to common.

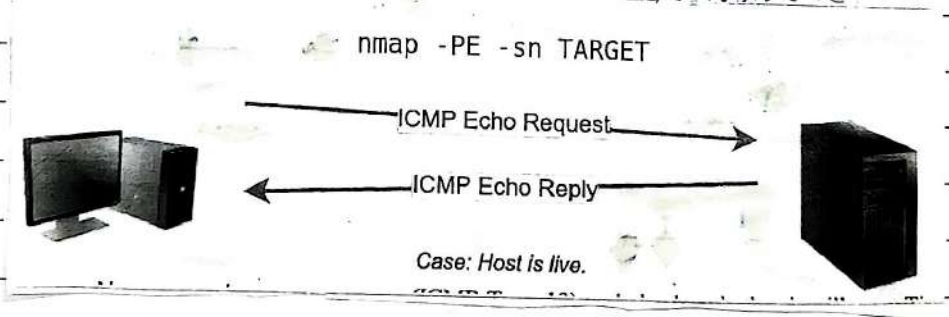
```
Pentester Terminal
pentester@TryHackMe$ sudo nmap -PR -sn 10.10.210.6/24

Starting Nmap 7.60 ( https://nmap.org ) at 2021-09-02 07:12 BST
Nmap scan report for ip-10-10-210-75.eu-west-1.compute.internal (10.10.210.75)
Host is up (0.00013s latency).
MAC Address: 02:89:75:8A:82:89 (Unknown)
Nmap scan report for ip-10-10-210-100.eu-west-1.compute.internal (10.10.210.100)
Host is up (-0.10s latency).
MAC Address: 02:63:00:18:2D:CD (Unknown)
Nmap scan report for ip-10-10-210-165.eu-west-1.compute.internal (10.10.210.165)
Host is up (0.00025s latency).
MAC Address: 02:59:79:4F:17:B7 (Unknown)
Nmap scan report for ip-10-10-210-6.eu-west-1.compute.internal (10.10.210.6)
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 3.12 seconds
```

In this scenario, the attackbox had the IP address 10.10.2016 and employed ARP request to identify, active hosts within the same subnet.

Nmap sends ARP request to all the traget machines and those that are online will respond with an ARP reply.

The ARP scan operates as depicted in the accompanying



From Computer 1.
To Computer 1 (to indicate it is broadcast)
Packet type : ARP request.

we will start by Configuring the routing the router.

Right-click on the router and select "CLI" this will open the Command.

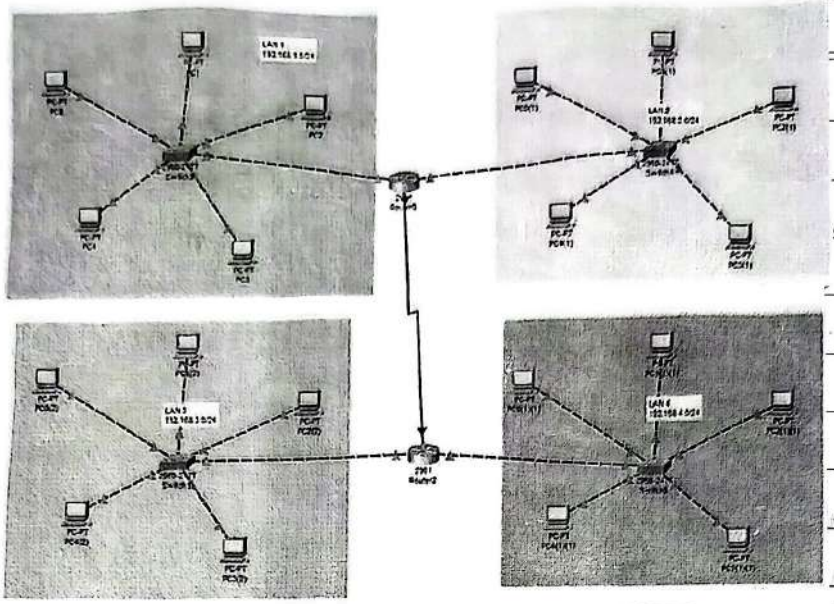
Testing the network:

now that our network topology is configured we can test the network, open a Command prompt on each Pc and try to ping the order Pc.

If the ping is successful, then the network is properly, we can also use the ping command to the Connectivity between the router and the Pcs.

output

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC4(2)	Router2	ICMP		0.000	N	12
	Successful	PC4(2)	PC2(1)(1)	ICMP		0.000	N	13
	Successful	PC0	Router0	ICMP		0.000	N	14



The IP addressing for the network shown in the topology can be as follows:

Router R1

PC1: 192.168.1.11
PC2: 192.168.1.12
PC3: 192.168.1.13
PC4: 192.168.1.14
PC5: 192.168.1.15

Fast Ethernet 0/2: 192.168.2.0/24

PC1: 192.168.2.11
PC2: 192.168.2.12
PC3: 192.168.2.13
PC4: 192.168.2.14
PC5: 192.168.2.15

Configuring the Devices:

Now that we have added our devices and connected them we can start Configuration.

Ex: 9.

SUBNETTING IN CISCO PACKET TRACKER

Aim:

Implementation of subnetting in cisco packet tracker simulator.

Classless IP subnetting is a technique that allows for more efficient use of IP address by allowing for subnet marks that are not just the default marks for each IP class. this means that we can divide our IP address space into smaller subnet.

Creating a network Topology:

The first step in implement classless IP subnetting to create a network topology in packet tracker. To create a network and Generic.

This will create a blank by dragging the we can use to add devices.

Adding the Devices:

Once we have created network topology we can add device to it. Here we will be adding routers, switches and PC's to add devices

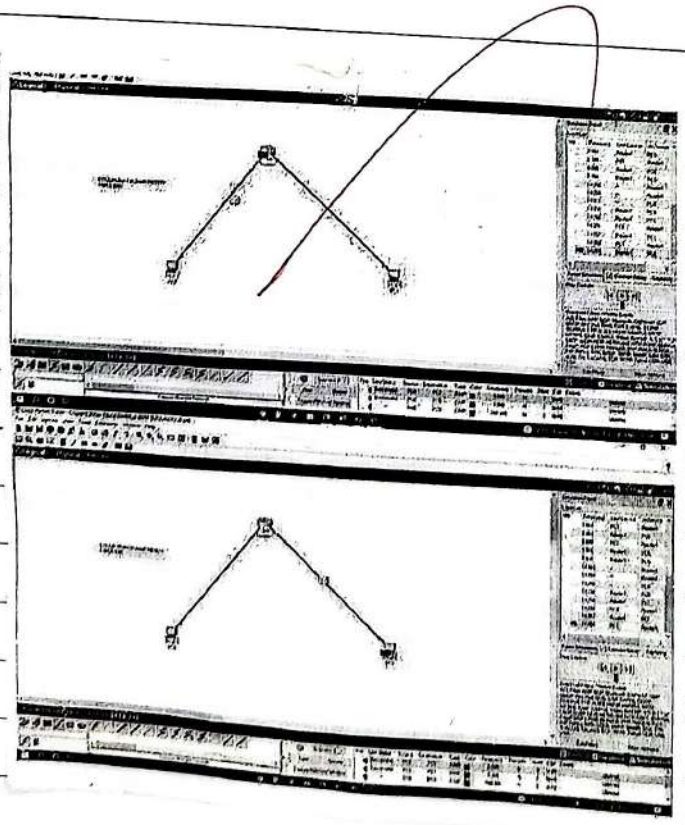
Select the device from the bottom in the top left corner.

Step 2 (Configuring PCs)

1) Assign IP address to every PC in the network.
2) Select the PC, Go to the desktop and select IP Configuration and assign an IP address
3) Assign the default gateway of PC0 as 192.168.10.1
4) Assign the default gateway of PC1 as 192.168.20.1

Step 3. (Connecting PC's with routers)

1) Connect fast Ethernet 0 port of PC0 with fast ethernet 0/0 port of router using a copper straight through cable



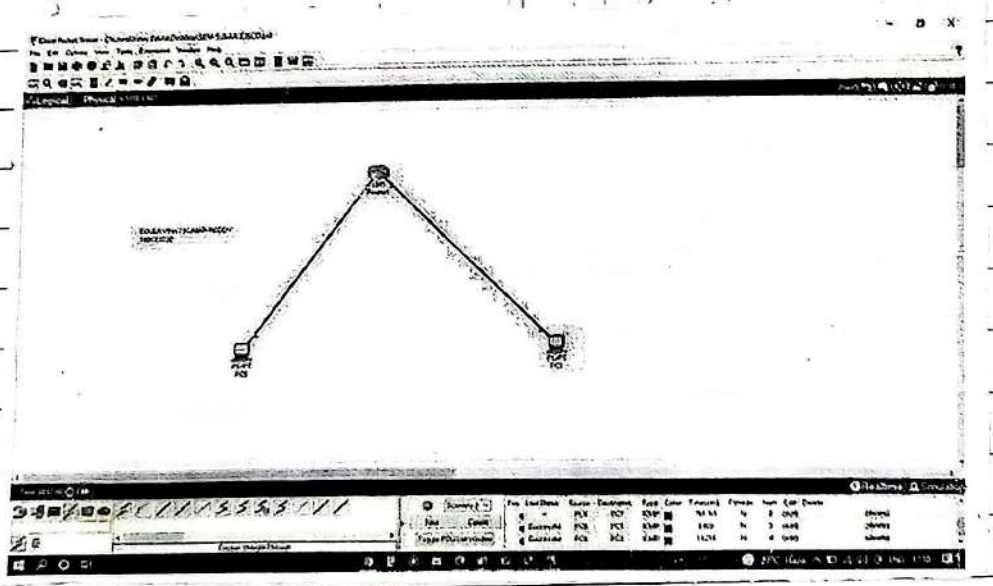
Ex-10

a) Internetworking with routers in Cisco packet Tracers

Aim:

Internetworking with routers in Cisco packet Tracers.

In this network a router and 2 PCs are used. Computers are connected with routers using a Copper Straight-Through cable. After forming the network to check network connectivity a simple PDU is transferred from PC0 and PC1



Procedure:

Step 1 (configuring Router1):

1) Select the router and open CLI
2) Press Enter to start.
3) Type enable to activate the privileged mode

Result

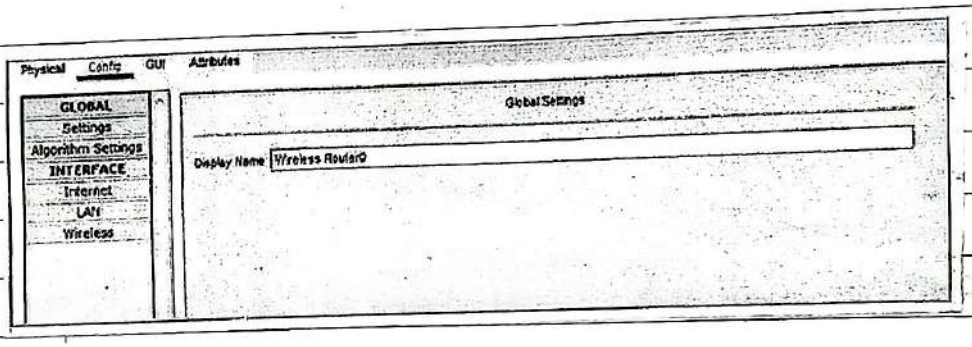
Thus the subnetting in Cisco packet Tracker was implemented successfully.

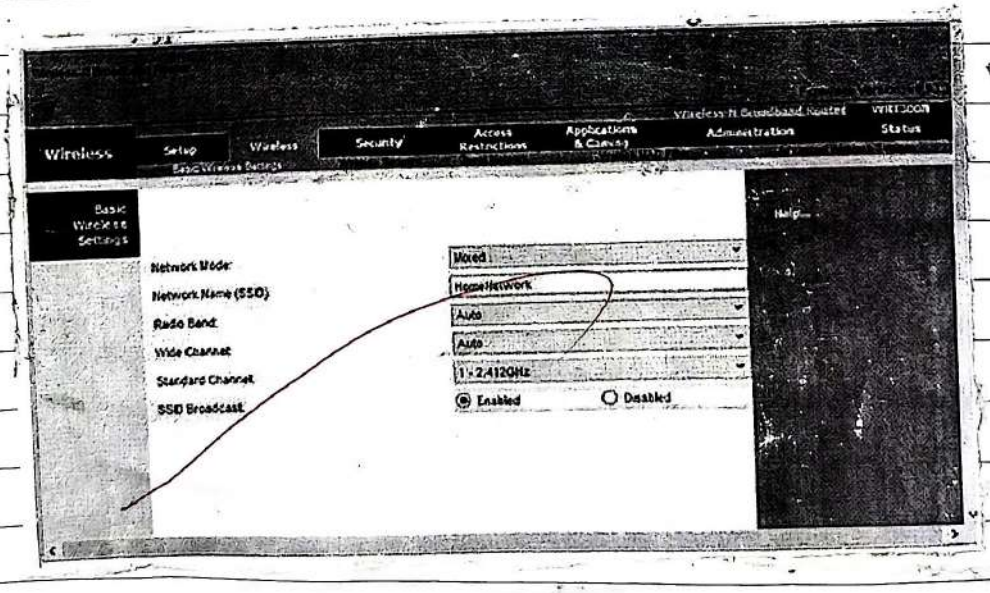
Part - 2 : Configure the network Devices

Create the wireless network on the wireless router.

Click on the wireless Router icon on the Packet tracer logical workspace to open the device configuration window.

In the window router configuration window, click on the GUI tab to view Configuration options for the wireless router.





Ex. 10 - b

Wireless router, DHCP

Aim:

Design and Configure an internetwork using Wireless router, DHCP Server and internet cloud.

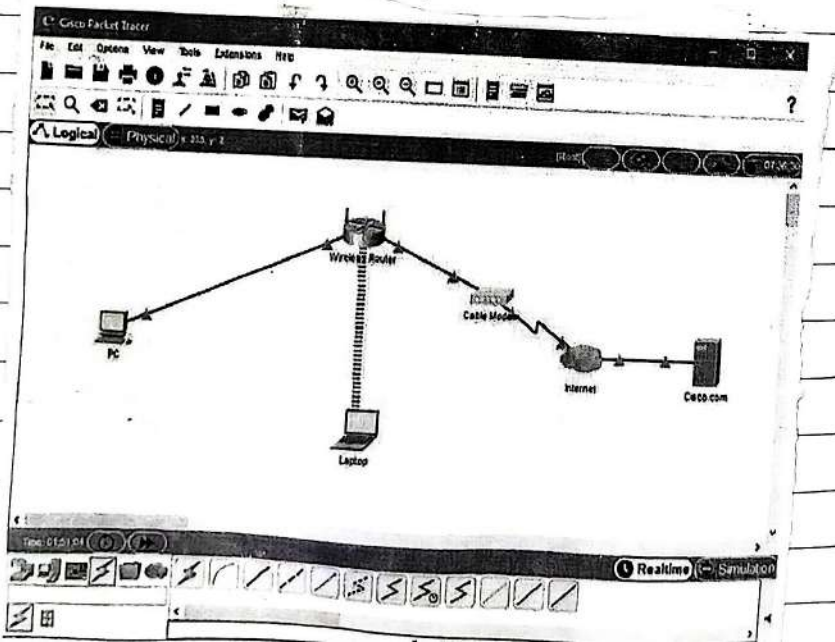
Part 1: Build a Simple network in the logical topology workspace

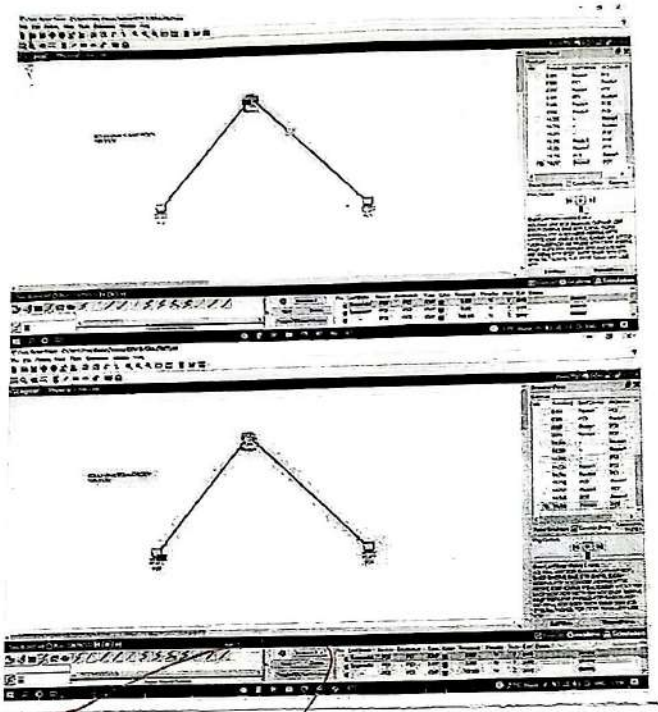
Step -1 Launch packet tracer

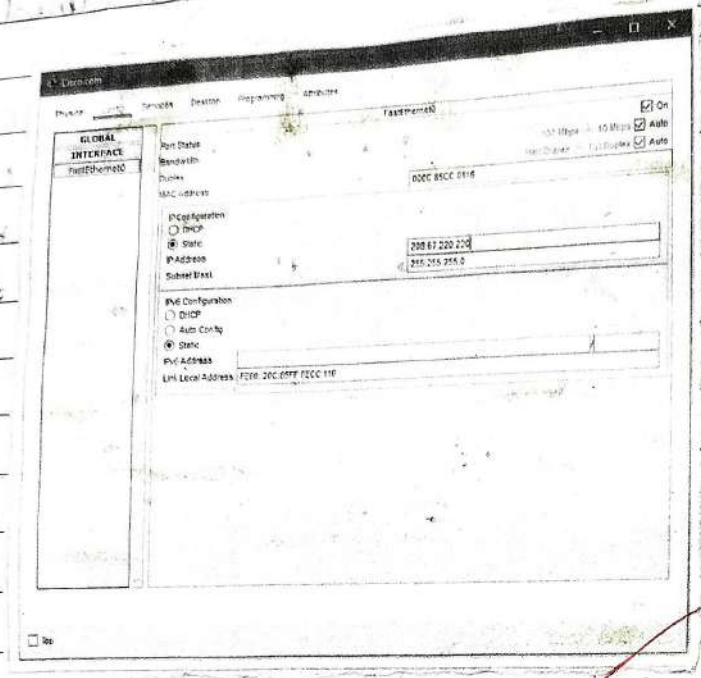
Step -2 Build the topology.

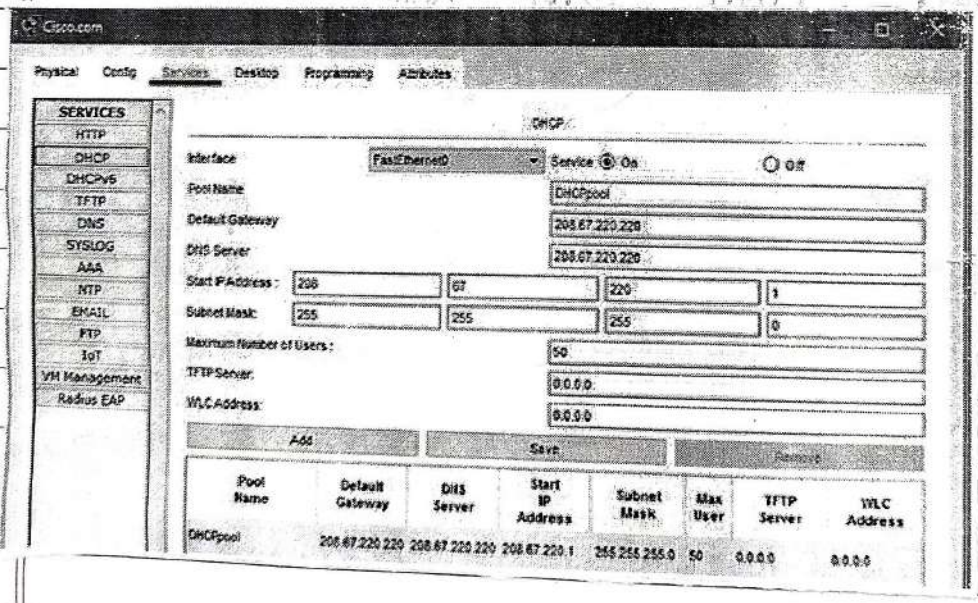
a) add network devices to the workshop

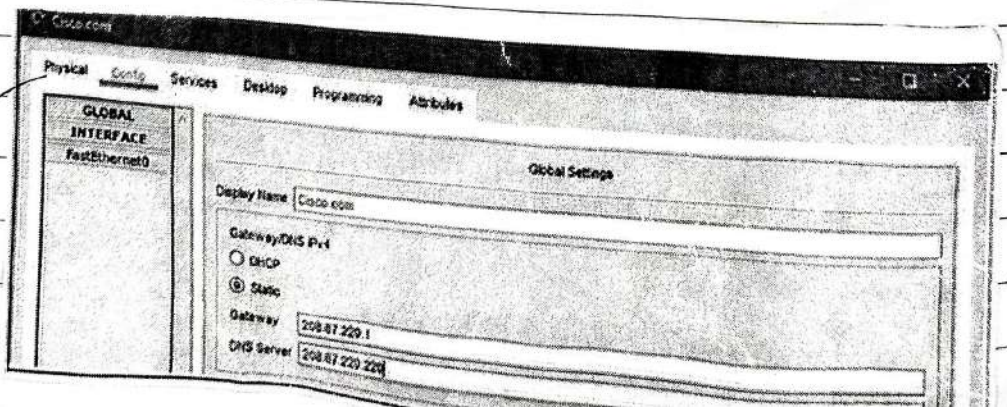
Using the device selection box, add the network devices to the workspace as shown in topology.






GLOBAL
INTERFACE
FastEthernet0
Port Status
Bandwidth
Duplex
MAC Address
IP Configuration
DHCP
Static
IP Address
Subnet Mask
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address
On
Auto


Cisco.com
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
DHCPpool
Default Gateway
208.67.220.220
DNS Server
208.67.220.220
Start IP Address :
208
67
220
1
Subnet Mask:
255
255
255
0
Maximum Number of Users :
50
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
DHCPpool
208.67.220.220
208.67.220.220
208.67.220.1
255.255.255.0
50
0.0.0.0
0.0.0.0

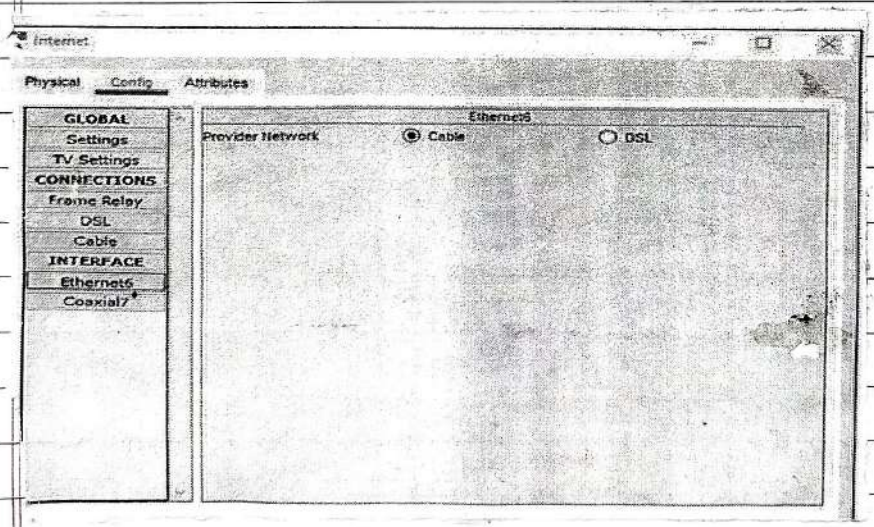

Cisco.com
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
INTERFACE
FastEthernet0
Global Settings
Display Name
Cisco.com
Gateway/DNS IPv4
DHCP
Static
Gateway
208.67.220.1
DNS Server
208.67.220.220

Step-4 Configure the Internal cloud.

Install network modules if necessary click on Internet cloud icon on the packet Tracer logical workspace and the click on the physical tab

Click on the config tab in the cloud device window. In the left pane click on cable under connections in first drop box.

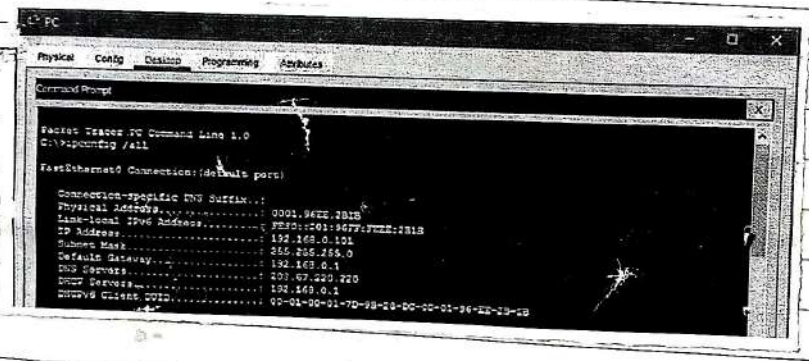
Ethernet then click the add button to add these as the from port and To port as shown in given figure you can see.

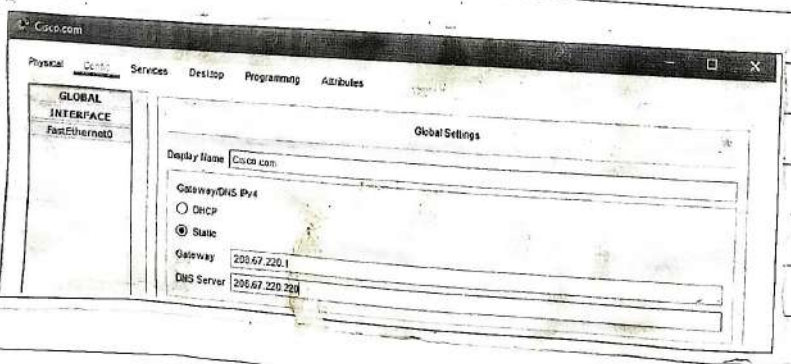


Step 3:- Configure the PC.

Configure the Pc for the wired network click on the Pc icon on the packet tracer logical workspace and select the Desktop tab and click then the IP Configuration window.

Use DCHP to receive an IP V4 address from the wireless router close IP Configuration window.





Ex: 11

Static Routing Configurating Using Cisco packet Tracer.

Aim:

Simulate static Routing Configurating Using Cisco packet Tracer.

static routes are the routes you manually add to the router's routing table. the process of adding static routes to the routing table is known as static routing. Let take a packet tracer example to Understand how to use static

In this lab, each network has two routes to each to will Configure one router as the main route and another route as the backup route

If you manully want to select a route that the router should add to the routing table you have to set the AD value of the route lower than other router.

ip route 30.0.0.0 255.00.0 20.0.0.2

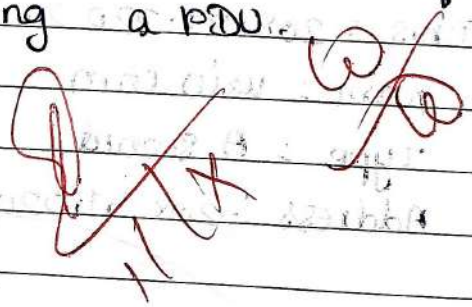
ip route 30.0.0.0 255.0.0.0 40.0.0.02 2

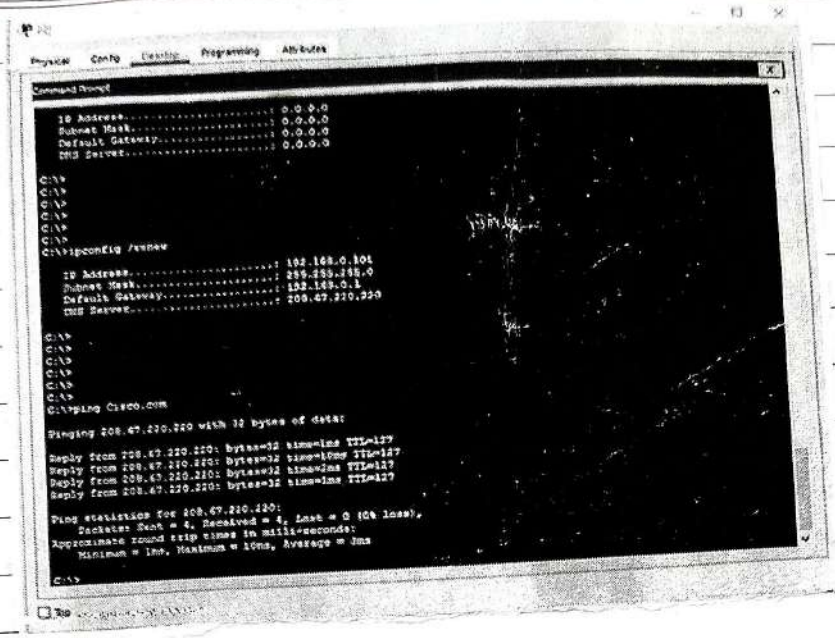
If the first route fails, the router automatically.

Date
Page

Result:

Thus a Simple internet work was designed Configured using a router in Cisco packet Tracer and connectivity between PC0 and PC1 was verified Successfull using a PDU.





Answer:

IP : 208.67.220.220

Gateway : 208.67.220.220

maximum user : 50

Dns : 208.67.220.20

name : usio.com

Type : A Record

Address : 208.67.220.220

Answer :

IP : 208.607.220.220

Subnet mask : 225.225.255.0

Gateway : 208.67.220.1

Dns : 208.67.220.220.

Use the 'show ip route static' Command to print all static routes

Note down the route you want to delete.

Simulate RIP using Cisco packet Tracer

Assign IP address to interface of routers

Result:

Thus static routing was Configured Successfully using cisco packet Tracer, and Communication between different network was verified.

router 1 requirement:

Create two router for network 10.0.0/
Configure the first route.

Create two routes for network 40.0.0/
and configue the first route.

Verify the router add only main routes.

Router 2:

Create static routes for network 10.0.0.0
and network 30.0.0.0/8 and verify
that router adds both routes to
the routing table

Verifying static routing

On router 0, we configured two
routes for network 3000.0/8 these rout
are are via router 1 and via router

We can verify this configuration in
two ways

By listing the routing tables entries
on router since a router table.

Creating, adding, verifying Static router.

Router automatically learn their Connected network. We only need to add routes for the network.

Router 0 requirements.

Create two routes for network 30.0.0.0/8. and configue the first route.

Create two router for the host 30.00.10 and configue the first route.

Verify that router adds only main router to the routing table.

Router 0 Configuration

Router > enable
Router # configue.
terminal
Enter configuration command, one per line
Router # ip router 30.0.0.0 255.000.210
Router # show ip route.
map 8 30.00.0/8 [10/0].

Input:

Enter a message from the client to the server.

output:

Enter message : Hello Server
message received from server : Hello Server

Result:

Thus the echo client-server program using TCP and UDP socket was implemented and executed.

Display the received message
Close the socket and stop the progr

UDP Echo client - Server:

Server:

Start the program
Create the UDP socket using Socket()
Blind the socket on IP address and port number.
Repeat the process until the program is Sendto()
Close the socket.

Client:

Start the program
Create a UDP socket
Enter a message to be sent to the server
Send and receive the message
Display the message
close the socket.

An echo server is a program that receive that a message from a client and sends.

the same message back. the client display the return message.

Ex-12

TCP / UDP Socket.

Aim

To implement an echo client-server programming using TCP / UDP Sockets.

Algorithm.

A TCP Echo client - server.

Server:

Start the programming
Create a TCP socket using Socket()
Bind the socket to an IP address and port number using bind()
Listen for incident clinet connection using listern()
Accept the client Connect using accept()
Send the received message back
close the Connection and stop the program.

Client:

Start the progeram
Create a TCP socket
Connect to the Server using its IP a and port number.
Read a number from the User.
Send the message from the server

Input:

Enter the IP address or hostname:

Enter the destination address:- 8.8.8.8

Output:

playing 8.8.8.8

Reply received from 8.8.8.8
Time =25ms

Rely: received from 8.8.8.8
Time= 28ms

packet sent: 2
packet received : 2
packert lost: 0

Result:

Thus the ping program was implemented Successfully to check network Connectivity and measure the response time of the destination Computer.

Input:

Enter the IP address or hostname:

Enter the destination address:- 8.8.8.8

Output:

playing 8.8.8.8

Reply received from 8.8.8.8
Time =25ms

Rely: received from 8.8.8.8
Time= 28ms

packet sent: 2
packet received : 2
packet lost: 0



Result:

Thus the ping program was implemented successfully to check network Connectivity and measure the response time of the destination Computer.

Ex-13.

Ping program.

Aim:

To implement the ping program to Check the connectivity between two Computer over a network

Algorithm:

Start the program.
Import the required network libraries
Get the ip address or hostname of the destination computer.
Create a socket to send and receive network
Send icm Echo Request packet to the destination.
wait for the icmp echo reply.
Calculate the response time
Display the reply message and respon time.
Display timeout message if no reply is received.

Repeat the process that required number of packets.
Stop the program.

Input:

Network packet transmitted received through the network interfa

output:

Packet Captured successfully

Source IP : 192.168.1.10
Destination IP : 192.168.1.1
Protocol : TCP

Packet captured Successfully

Source IP : 192.168.1.10
Destination : 8.8.8.8
protocol : UDP

Result:

thus packet sniffing using raw Socket was implemented successfully and capture the packet detail were displayed

Ex-14

RAW Socket.

Aim:

To write a program using RAW Socket to implement packet sniffing

Algorithm:

Start the programming
Import the required socket library
Create a raw socket to capture network packet
Bind the socket to the appropriate network interface if required
React Pact using packet
Extract the IP header information from the Capture packet
Identify the source IP address a destination IP address

Display Captured packet details on the Screen.
Continue capturing packet Until th program is terminated.
Close the Socket and stop the program.

Input :

Enter the IP address or hostname :

Enter the destination address :- 8.8.8.8

output :

playing 8.8.8.8

Reply received from 8.8.8.8
Time =25ms

Rely: received from 8.8.8.8
Time= 28ms

packet send: 2
packet received : 2
packert lost : 0

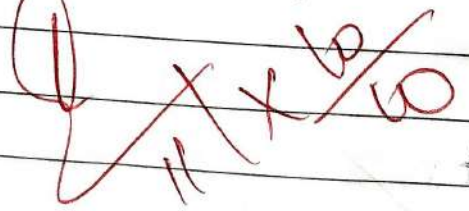


Result :

Thus the ping program was implemented Successfully to check network Connectivity and measure the response time of the destination Computer.

Result:

Thus the different types of web logs were analysed successfully using the Webalizer tool.



Introduction to an airline

Ex-15

Web Logs

Aim:

To analyze the different type web logs using the webalizer tool

Procedure:

Run the webalizer window version

Download and select the web log file (access_log)

specify the target directory for the output file

Click the run webalizer option to analyze the log file

View the generated reports and graph

Analyze the statics, including websi visits request, URL's and Traffics.

Input:

Web server log file (access log).

Output:

Webalizer generates reports and graphical statistics showing website using visitor information daily and montly Request.

URL's accessed and data transfered